GROUP INSIGHT REPORT

Christchurch International Airport Limited

Sustainability Tour · 18 June 2026

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Photography by Kai J. Lee

Executive Summary

On 18 June 2026, twenty-one sustainability practitioners from across New Zealand, the UK, the US, Thailand, Poland, and Cambodia joined Christchurch Airport for a behind-the-scenes tour of the airport's sustainability operations. The tour covered airside infrastructure, the food court and terminal, and the waste sorting room, followed by a boardroom debrief.

Headline findings:

1. CIAL has achieved a verified 92% reduction in Scope 1 and 2 emissions since 2015, an uncommonly deep result for a regional airport, supported by a 170 MWdc solar farm, a closed-loop ground source heat pump, and full fleet electrification.

2. The waste team's back-of-house sorting operation is the only intervention that has measurably improved waste diversion at the airport. According to CIAL's experience, consistent with international airport research, public-facing bin trials have consistently failed to shift passenger behaviour.

3. The PDX Green Plate model from Portland International Airport offers a credible operational template for reusable serviceware in the food court. CIAL has a space adjacent to the food court that could support centralised dishwashing, and Shannon's team has already proposed that the airport own a reusable fleet, cups included. The critical path is not infrastructure but tenant adoption, which requires a structured behaviour change strategy built around understanding and pre-empting operator resistance.

4. Participants consistently valued the honesty with which CIAL shared both achievements and failures, from the "climate positive" story to the autonomous shuttle that found no use case.

Tour Context

Christchurch International Airport is the gateway to the South Island. It is 75% owned by Christchurch City Council via Christchurch City Holdings Limited. The airport serves approximately 7 million passengers annually and operates a campus of over 1,400 hectares, including the Kowhai Park renewable energy precinct.

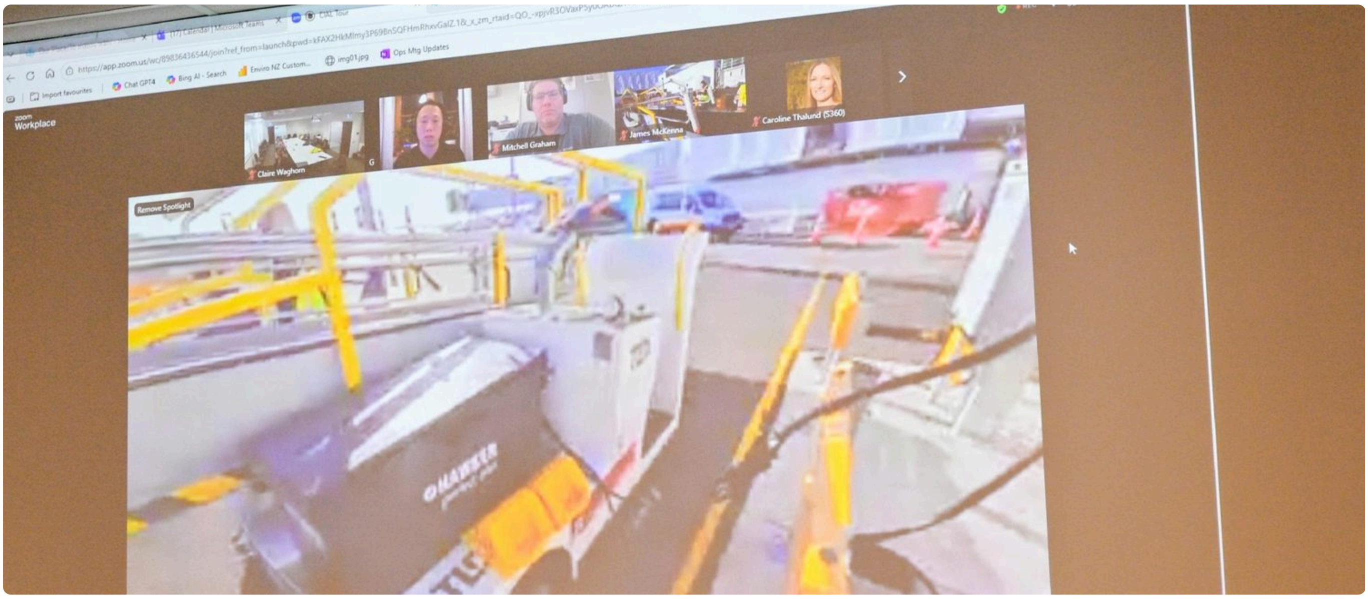
The tour visited:

Airside	Ground power units, EV charging hubs, electric fire truck, fuel infrastructure, quarantine waste handling, air bridge wheels, aviation security confiscation cabinets
Terminal	Food court (SSP-operated outlets, Fush, Koru Lounge), sustainability wall, Method bins, refurbished furniture
Waste room	Manual sorting table, BI-connected weighing system, food rescue area, Flybrary book salvage, organic bins, loading dock recycling

Boardroom	Debrief on communications, stakeholder relations, storytelling, and the "climate positive" experience

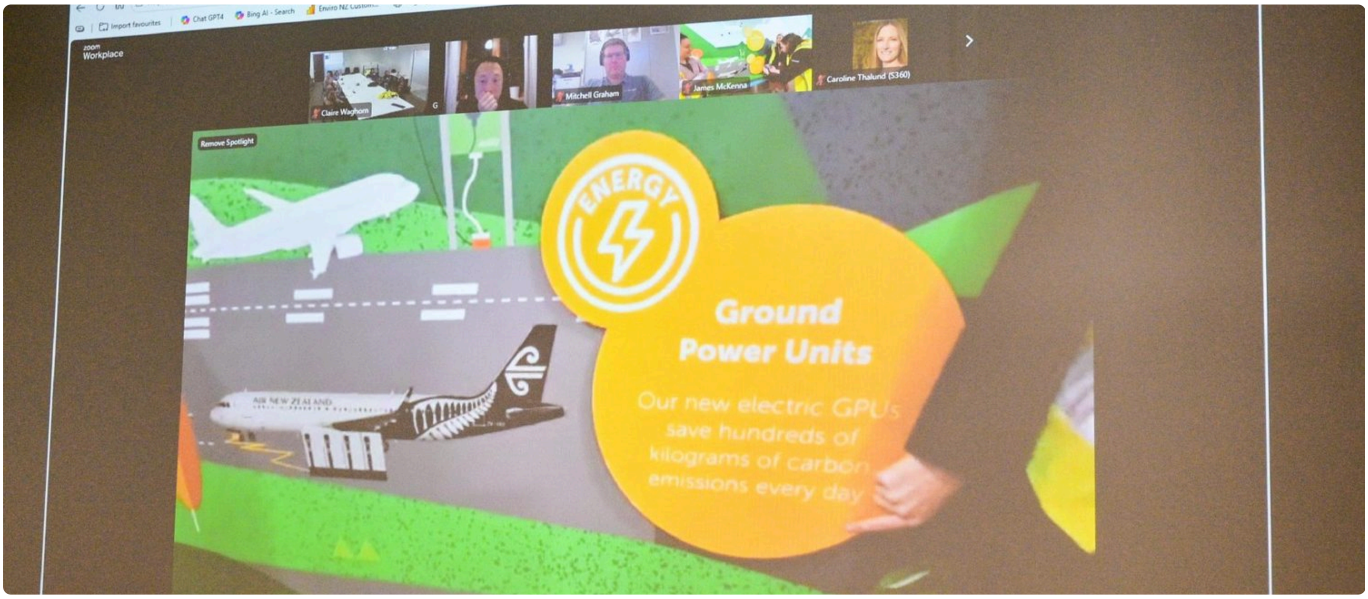
Key Observations

Energy and Decarbonisation



CIAL's emissions reduction story is substantial and verified. The airport has reduced its direct emissions (Scope 1 and 2) by 92% from a 2015 baseline, with a target of absolute zero by 2035. The remaining emissions come from a small number of utility vehicles, most of which have already been transitioned.

The infrastructure behind this is visible: planes plug into ground power units instead of burning auxiliary fuel; ground service equipment charges at EV hubs whose electrical pipes were deliberately specified during the terminal's construction; a closed-loop ground source heat pump heats and cools the terminal without contaminating the aquifer; and the 170 MWdc Kowhai Park solar farm, one of New Zealand's largest, is nearing operational completion.



A detail that stood out: other New Zealand airports have struggled to electrify their ground service equipment because they lack the charging infrastructure. CIAL has capacity to spare, because EV charging infrastructure was deliberately included in the terminal design. This kind of long-horizon thinking, bundling infrastructure costs into airline charges negotiated eight years in advance, is a model other airports could learn from.

The electric fire truck, the first in the Southern Hemisphere, drew attention both for its capability (full battery capacity for sustained firefighting, not scaled down despite a three-minute maximum drive time) and for the public backlash it attracted locally.

One structural barrier to faster decarbonisation at airports is the concentration of fuel supply infrastructure. Where a single provider controls the ground-based fuel system, the airport has limited leverage to accelerate alternative fuel adoption regardless of its own commitments. This is not unique to Christchurch. It is a sector-wide constraint that warrants regulatory attention.

Waste and Circularity

The waste story is where CIAL is most honest about its struggles. According to CIAL's experience, consistent with international airport research, the airport has trialled a wide range of public bin configurations, including different sizes, messaging, number of streams, and designs, and nothing has produced measurable improvement in passenger sorting behaviour.

What does work is the back-of-house sorting table. Two EnviroNZ employees sort through landfill waste six hours a day, six days a week, pulling recyclable material out of the general waste stream. The operation is weighed and categorised through a BI-team-built tablet system that produces detailed waste composition data. This data has been shared with Heathrow and other airports as a pilot scheme.

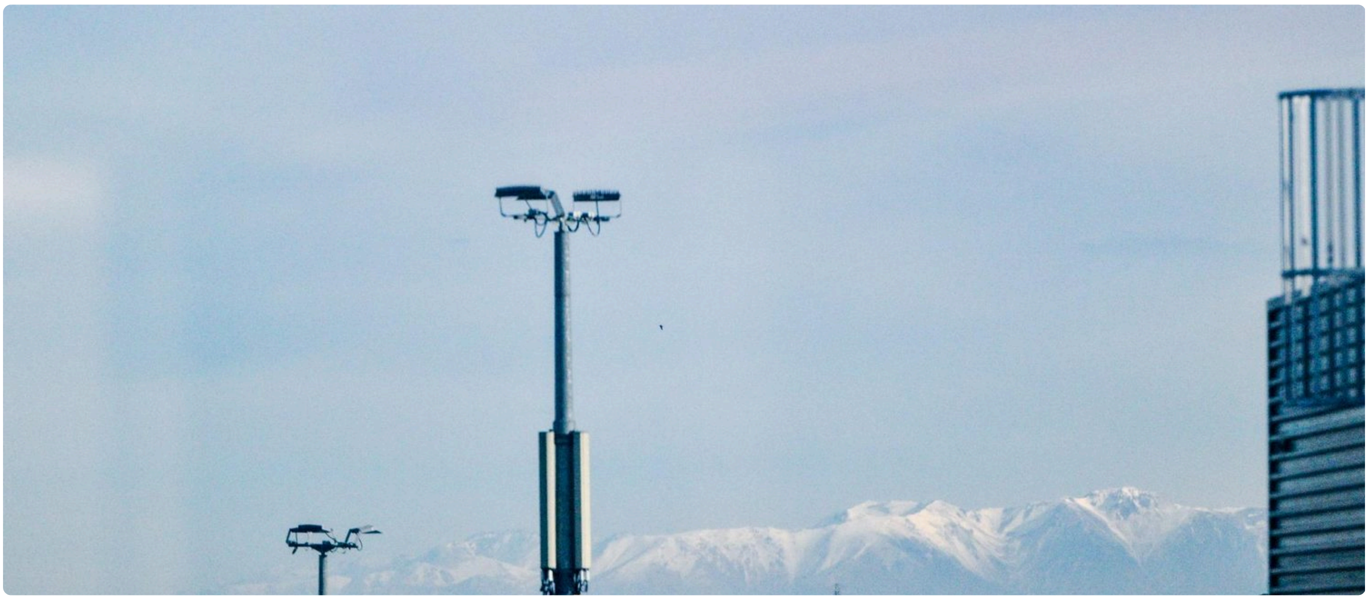
This finding is also consistent with research conducted with Christchurch City Council on the Circular Serviceware Opportunity in Christchurch, carried out in 2024 and 2025. This research is available at no cost on request from the tour team.

The current diversion rate is just over 50%, against a target of 80%. Research and field experience from comparable sorting operations suggests that 50% may represent a structural ceiling when organisations rely primarily on sorting and recycling infrastructure without adopting circular processes upstream. Sorting lines are necessary but not sufficient. The path beyond 50% requires redesigning what enters the waste stream in the first place, not just improving what we do with it once it arrives.

In the food court, roughly half of the outlets serve on crockery and half are takeaway-only. Shannon's team is pushing toward reusable packaging, and the airport is considering owning a fleet of reusable serviceware rather than asking each tenant to manage their own.

A small but meaningful detail: Coso and the waste team rescue unopened confiscated food for the city food bank (3 tonnes this year). This approach mirrors food recovery work at Venues Otautahi, Christchurch's major events venues, which established a similar programme around 2023. The team is also building the Flybrary, the first airport library in New Zealand, stocked entirely with books salvaged from the waste stream.

Biodiversity and Land



The airport sits on a large campus with ecologically significant dry land habitat, a 200-year-old Kowhai tree (now in the middle of the solar farm), and legacy landfill pockets requiring remediation. The Avenex grass story was particularly honest: a monoculture developed with Lincoln University to repel birds (a genuine safety requirement near runways), once seen as a win, now recognised as bad for biodiversity and soil health. CIAL wants to replace it.

Communications and Community

The most emotionally resonant part of the tour was the "climate positive" story. CIAL achieved certified climate positive status (120% offsets in New Zealand native forestry), intended to keep it as an internal achievement, saw it publicly announced by the then-Prime Minister, then watched it be perceived as greenwashing despite extensive caveats. Claire pulled the claim from all materials before an Advertising Standards Authority complaint arrived, which was ultimately not upheld due to CIAL's proactive actions.

This experience shaped the airport's current approach: telling honest, challenge-first stories rather than polished narratives. As Claire put it, "My favourite stories are the bad news stories. They're received quite well."

Post-Tour Research Findings

Our post-tour research identified several findings directly relevant to the challenges raised during the tour. We offer these in the spirit of the collaborative, honest exchange that characterised the day.

1 Reusable serviceware works when it is paired with a tenant adoption strategy

Reusable serviceware addresses dine-in waste, but a significant volume of food court waste comes from takeaway cups. The instinct is to solve this with a dedicated returnable cup system platforms like Again Again operate across New Zealand hospitality. But if CIAL owns the reusable fleet and washes it centrally, cups belong in that fleet alongside plates and cutlery. The system is the same; the cup is just another item in the loop.

The harder problem is tenant adoption. SSP operates the majority of food court outlets and CIAL cannot mandate participation unilaterally. Experience from PDX's Green Plate Program and from circular serviceware research conducted in Christchurch in 2024–2025 confirms that reusable systems succeed or fail on operator buy-in, not on logistics.

This means the rollout plan should be paired with a behaviour change strategy refined through an accusation audit - identifying every objection tenants are likely to raise (hygiene concerns, speed of service, breakage costs, customer confusion) and addressing them before the first conversation. A resistance map built using the 5 Spheres of Empathy framework can structure this work systematically across both direct stakeholders (outlet managers, front-of-house staff) and indirect stakeholders (SSP regional management, facilities contractors). The prompt chain for this process is available at https://5spheresofempathy.com/resources, which together with the circular serviceware research conducted in Christchurch can help with de-risking the strategy.

Amsterdam Schiphol offers a useful reference point. The airport trialled reusable cups in its Lounge 4 and reported success in April 2026. Schiphol Airport Retail also ceased selling packaged plastic water bottles across its twelve shops, removing an estimated 750,000 single-use bottles per year from the terminal waste stream. The evidence confirms reusable systems can operate in airport environments. What it does not yet provide is published adoption rate data - which is why the behaviour change strategy matters more than the system design.

2 The PDX Green Plate model provides a credible operational template for CIAL's existing dishwashing room

Portland International Airport (PDX) launched the Green Plate Program after discovering that 75% of food court customers ate in but were being served on takeaway packaging anyway. By switching the default to reusable plates and cutlery (takeaway packaging on request only), the pilot achieved a 73% reduction in to-go packaging waste and reused 10,000 dishes in three months. The program relaunched in December 2025 and is in early stages of its current deployment, with a target of diverting 15 tonnes of waste annually.

Further operational detail on an adapted version of this model is available from the Circular Serviceware research conducted in Christchurch in 2024 and 2025, available at no cost on request from the tour team.

The operational model is simple: the airport owns the plates and cutlery. The airport's maintenance contractor collects and washes them in a centralised dishwashing room. A commercial Energy Star dishwasher completes a cycle in three minutes using one gallon of water. Tenants serve food on reusable serviceware as default.

CIAL has a space adjacent to the food court that could support centralised dishwashing. Shannon has already proposed that the airport own a reusable fleet. SSP's global dine-in-first policy aligns with this approach. The remaining steps involve confirming the fit-out requirements for that space and developing the tenant engagement strategy needed to bring operators into the system.

3 Back-of-house sorting is global best practice, not a workaround

The tour group saw the waste sorting table and may have interpreted it as a stopgap. It is not. Research across airports globally, including Schiphol's Baseline Circular Airports Method, confirms that back-of-house sorting is the most effective intervention for airport waste diversion. Public bin design, signage, and education campaigns have consistently underperformed at airports that have tried them.

That said, two low-cost experiments are worth running:

- Transparent bins (field experiments run by the tour facilitator at large-scale food festivals in Cambodia, where transparent bins significantly outperformed standard bins with black bags and iconography, including in combination with volunteer staffing at bin stations): making bin contents visible creates social proof and has been effective in high-volume public settings. These bins are not aesthetically elegant. Field evidence suggests they work regardless. A short trial with CIAL's brand team involved is worth the conversation.
- Push-lid bins (Mitchell's theory): requiring a deliberate opening action adds friction that encourages a moment of thought. CIAL has not tested this.

There is also an upstream opportunity that the 50 percent diversion rate does not yet capture. Diverting material before it reaches the sorting line, potentially through passenger-facing systems tied to community milestones, could open a new chapter in how the airport communicates its waste story publicly and builds shared ownership of the goal with the people moving through it.

4 Air bridge wheel recycling has NZ options

Claire asked the group for ideas about recycling the rubber-bonded-to-steel air bridge wheels. Research identified:

- Treadlite (Cambridge, NZ), the country's sole tyre recycling plant, processes 9,000+ tyres per week and handles solid tyres. They should be contacted to confirm they can accept solid bonded industrial wheels.
- Tyrewise, New Zealand's product stewardship scheme (launched March 2024), now covers solid tyres for motorised vehicles and exceeded its first-year collection targets. CIAL may already be familiar with Tyrewise. The immediate action is confirming whether air bridge wheels fall within the scheme's current scope, and if not, whether they can be added.
- Trelleborg, the OEM supplier of passenger boarding bridge tyres, should be asked about take-back or recoring arrangements.
- REVYRE (Energy Estate + InfraCo), an AU/NZ joint venture using Tyromer devulcanisation technology. REVYRE appears to already have a presence in North Canterbury. Confirming their current capability and capacity for solid rubber recycling should be treated as a near-term action rather than a longer-term watch item.

5 Airside heat loss through vehicle openings

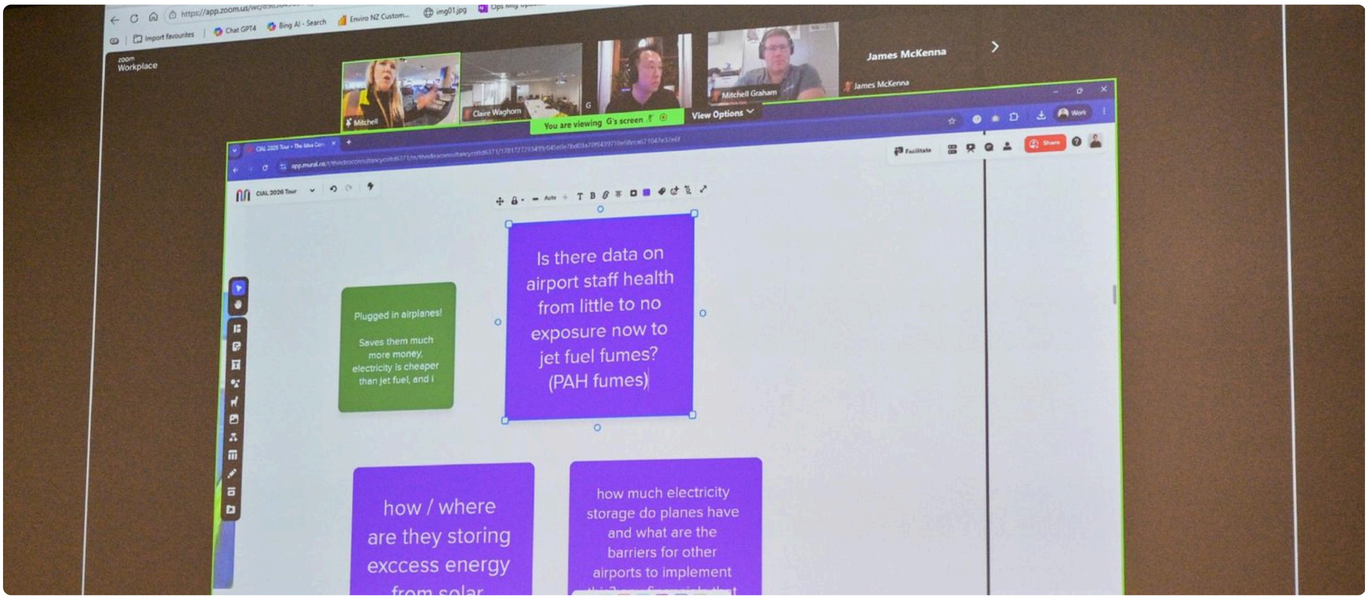
During the airside portion of the tour, participants observed heated air escaping through large vehicle access openings, the points where ground service equipment moves between the building interior and the apron. In a Christchurch climate with Canterbury nor'westers and cold southerlies, every unit of heat lost through an open door must be replaced by the ground source heat pump, adding electrical load to a system already doing the work of heating and cooling the terminal.

In researching this observation, we found that most published material on airport heat loss is vendor marketing rather than independent research. The one peer-reviewed study we identified - a field programme across eighteen airport terminals published in Energy (Elsevier) - found that air infiltration through openings accounted for 18–71% of total heat loss in terminal buildings, with airtightness improvements and radiant floor heating achieving an average 84% reduction in annual heating demand. That study addresses terminal-scale infiltration rather than vehicle access points specifically, but the underlying physics are the same: unprotected openings in heated buildings lose conditioned air rapidly, and the losses compound in windy or cold conditions.

Established interventions in large-building operations include air curtains or air barriers across frequently opened doors, high-speed doors that minimise exposure time during vehicle transit, and destratification systems in high-ceiling spaces that push warm air back down to the working level. For openings that cannot be closed, radiant heating outperforms forced-air alternatives because it heats people and surfaces rather than air volume, remaining effective even when doors are open.

CIAL will likely have considered some or all of these options already. The observation is included here because the tour participants noticed it independently, and the research gap - very little rigorous, non-commercial work on heat loss through airside vehicle openings - may itself be worth noting.

Extended Findings



This section captures six themes that emerged during the tour but were not covered in the main body of this report. They are drawn from a rescan of the full transcript and Mural board contributions, supported by post-tour research.

Educating the Regulator

During the airside tour, Claire described a dependency that rarely features in airport sustainability conversations: the regulatory environment. New Zealand's Civil Aviation Authority has established an emerging technologies team to begin certifying novel propulsion systems, including electric, hydrogen, and hybrid, but the regulatory timeline does not yet match the pace of technology development. As Claire put it, CIAL needs to "line up with the regulatory side as well as the airlines as well as the aircraft manufacturers and then the energy side." In the debrief, she identified the risk more directly: "the regulatory system won't be fast enough."

This raised a question on the Mural board that deserves attention: "What can private sector do to help educate the regulator?"

There are strong international precedents for this. The UK Civil Aviation Authority has appointed an Environmental Sustainability Panel of eight external experts who act as a "critical friend" to the regulator, contributing approximately one day per month. The FAA's CLEEN Program has partnered with Boeing, GE, Rolls-Royce, and others across four phases since 2010, with total investment exceeding USD 500 million and a fifth phase initiating in 2026. And in New Zealand, the CAA's own Reduced and Zero Emissions Project (RZEP) is actively seeking industry partners. CAA Director Kane Patena has stated that "emerging aviation technologies often don't fit within existing regulations, so our job is to facilitate the safe integration of these innovations."

CIAL is likely already engaged with the NZ CAA's RZEP initiative. One opportunity that emerged from the tour discussion: a lightweight, informal channel where sustainability practitioners and CAA colleagues can share emerging technology and regulatory intelligence in near-real time. Email and LinkedIn move too slowly for a fast-moving technology environment. A simple instant messaging group or a brief regular update, something that takes minutes to read and seconds to contribute to, could fill a gap that currently has no owner. Whether CIAL is the right organisation to convene this, or whether it emerges from a broader coalition of practitioners, is worth exploring.

Shared Goals, Different Metrics

> **OBSERVER NOTE FROM THE TOUR TEAM**
>
> This theme is recorded as a facilitator observation. It reflects what was said on the tour and captures a dynamic that participants and hosts openly discussed. It is not an external judgement on CIAL's internal structure.

A tension that will be familiar to many sustainability leaders emerged in the tour debrief: when commercial and sustainability teams operate within different performance frameworks, decisions about the physical environment can become contested territory.

This is not a values conflict. Both teams are working toward the airport's success. The friction arises when short-term revenue optimisation and long-term sustainability targets pull in different directions, and there is no shared metric that holds both accountable simultaneously.

The practical implication is that progress in this area is unlikely to come from persuasion alone. It requires a governance question: what would a shared performance framework look like that gives both teams a reason to move in the same direction?

The tour team's observation is that this structural tension is among the most common failure points in institutional sustainability programmes, and it rarely appears on the risk register.

Sustainability Wall Placement and Guerrilla Communications

Claire identified the placement of CIAL's sustainability journey wall as a "pet peeve": "This is actually a place, it's a hallway on the way to the Koru Lounge so this is not everybody that walks through here, this is just Koru Lounge people that walk through here. That's a pet peeve of mine. I would love this to be more publicly accessible."

Mr. G noted this as a red sticky and asked whether there is data anywhere on the impact and memorability of sustainability messaging in airports. The answer is yes, and the evidence strongly favours moving messaging away from low-traffic corridors and into locations where people have time and few distractions.

Three findings from the research are particularly relevant:

- Copenhagen's "Pure Love" green footsteps reduced street litter by 46%. Green footprints painted on pavements led pedestrians toward correctly labelled bins. The initiative won Copenhagen the title of "cleanest city in Europe" and remains the most frequently cited example of visual pathway nudging delivering a measurable behavioural outcome.
- Toilet door advertising achieves 78-90% recall rates. People are a captive audience in bathrooms for 1.5 to 2 minutes, longer in parent rooms. Poster frames on the back of toilet doors are already used routinely for helplines and safety messaging. At CIAL, this is likely the highest-impact, lowest-cost channel available for sustainability communications, and it reaches every passenger regardless of which corridor they walk down.
- Eye-level bin signage outperforms waist-level labels. Airport-specific research found that bin labels displayed at waist level or below are significantly less likely to be read, particularly problematic in international terminals where language-agnostic visual design is critical.

Claire has already demonstrated a willingness to act on messaging instincts. She described physically removing the "climate positive" claim from walls when it no longer felt right. The sustainability wall content is strong. The problem is entirely one of placement and channel.

Legacy Landfill: Lessons from the Ground Up

Claire mentioned that pockets of underground landfill are scattered across the CIAL campus, the legacy of previous generations who "thought it was a really good idea to dig holes and put waste in it." Whenever these pockets are encountered during construction or development, they require remediation: excavation, assessment, and safe disposal at today's costs for yesterday's decisions.

Claire's own phrasing, "previous great ideas," captures the irony precisely. This is not primarily a CIAL operations issue. It is a lesson about the cost of getting it wrong the first time, paid by the people who come after you, usually decades later, usually at a multiple of what it would have cost to do it properly.

CIAL's remediation experience represents a teaching asset that very few organisations possess. The financial and environmental cost of past decisions is not abstract here, it is visible, measurable, and ongoing. One opportunity worth exploring: could CIAL offer this as a case study resource for Canterbury council policymakers, urban planners, or infrastructure developers? A site visit, a short case study document, or even a micro-course hosted on the CIAL website could make this story available to the decision-makers who most need to hear it.

Research into whether other airports have done this produced limited examples, which suggests CIAL could be a genuine first mover in turning remediation experience into policymaker education. The closest parallel is the Queen Elizabeth Olympic Park in London, where the London Legacy Development Corporation transformed 110 hectares of heavily contaminated brownfield into a regeneration model, then built a formal knowledge-sharing programme around the experience. The LLDC offers specialist guided tours for professional groups including urban planners and government officials, publishes detailed governance and remediation documentation as a public resource, and partners with universities including UCL for teaching. The Olympic Park remediation involved 2.2 million cubic metres of excavated soil, 80% of which was treated and reused on-site. While that scale dwarfs CIAL's challenge, the principle is the same: turning a liability into a contribution. Canterbury has a strong education sector and an active regional council. The audience for this kind of resource already exists locally.

Fush, Local Tenants, and the Chain Gap

The food court tour revealed a contrast between two types of tenants that is worth drawing out. On one side, the majority of outlets are operated by SSP, a global food service company with sustainability commitments at the global level. On the other side sits Fush, a locally owned fish and chip shop run by Anton Matthews, described by both Claire and Mr. G as "a Christchurch legend." Fush incorporates Te Ao Maori into its operations, has genuine community impact, and represents the kind of tenant that makes an airport feel like it belongs to its city. Claire responded warmly to the idea of doing a "Sunday dinner" educational advertising piece featuring Fush and its sustainability story.

Marika Vallier noted during the tour and on the Mural board a question that gets to the heart of this: "Are these companies shouted out to?", meaning, does CIAL publicly celebrate its sustainable local tenants?

The San Diego International Airport Green Concessions Program offers a model for doing exactly this. SAN operates a tiered certification system where concessions earn points for actions such as using green cleaning supplies, selling reusable products, and training staff. By early 2020, 75% of concessions were certified "SAN Green Concessions." The program is recognised by the US EPA as a national case study. SAN embeds sustainability expectations into its operating framework and provides recognition, marketing support, and public visibility as the incentive.

This approach is gaining traction internationally. Unibail-Rodamco-Westfield, the global shopping centre operator, has developed a Sustainable Retail Index in partnership with Good On You and WWF France that scores approximately 1,200 brands across 4,300 stores on sustainability performance. The index is integrated into leasing decisions: higher-scoring tenants are favoured in tenant mix curation and marketing. URW has extended this model to airport retail at major hubs including LAX, JFK, and Chicago O'Hare. Separately, the Port of Seattle has incorporated legally binding green lease provisions into standard lease agreements for both airport and waterfront tenants, requiring specific environmental practices tailored to each tenant's context, supported by ongoing technical assistance. In Australia, over 60% of commercial leases signed in Sydney's CBD now contain green clauses, a fourfold increase since 2009, driven by the Better Buildings Partnership and national policy.

A CIAL version of this model could serve two purposes: it would create a structure for holding SSP accountable to its own global commitments at the local level, and it would give tenants like Fush the public recognition they deserve. Shannon has already spoken with the San Diego team. The groundwork is done.

Tenant Mandates, Penalties, and Rewards

> **OBSERVER NOTE FROM THE TOUR TEAM**
>
> This theme is recorded as a facilitator observation. It captures a policy conversation that took place during the tour debrief. The perspectives shared were offered openly by participants and hosts as part of the group discussion.

One of the most substantive policy exchanges of the tour centred on whether CIAL should mandate, penalise, or reward tenants on their waste journey. The tour facilitator raised the question directly, noting that food court tenants operate in a high-traffic, mid-to-high-income environment, and drawing a parallel to a wider pattern: "There is a cultural barrier, and it's back to people again to enforce that. We're so afraid to use the stick in New Zealand."

Shannon favours a collaborative approach, citing her conversation with San Diego International Airport, where the waste team found that working alongside tenants produced better long-term outcomes than enforcement alone.

Barry Lynch from Nezo provided a counter-example from his NZ and Australian operations. He described a customer whose business would never have engaged without a policy mandate: "Policy and compliance is a massive driver." He extended the point to forestry, where existing legislation is not being enforced due to fear of economic disruption, rather than thinking about the opportunities new industries could create.

The tension between Shannon's preference for collaboration and Barry's evidence that mandates work is not a disagreement. It is the same conversation happening in every jurisdiction in New Zealand. The cultural reluctance to enforce is real. So is the evidence that enforcement drives adoption. San Diego's model, which CIAL is already studying, suggests the answer may be both: set clear expectations in how the airport operates, but invest in the relationship and recognition infrastructure that makes compliance feel like partnership rather than punishment.

Questions Explored

WHAT WE ASKED	WHAT WE FOUND	CONFIDENCE
Has anything worked for public waste sorting at airports?	No. Global evidence confirms that no public-facing bin intervention reliably improves passenger sorting. Back-of-house sorting is the effective approach.	High
Can the food court shift to reusable serviceware?	Yes. PDX's Green Plate Program proved the model. CIAL has a space that could support centralised dishwashing and has proposed owning a reusable fleet. The main barrier is capital investment and ongoing servicing costs, followed by tenant adoption, which requires a structured engagement strategy rather than a technical solution	High
How did CIAL get EV infrastructure approved at design phase?	Costs bundled into airline charges negotiated 8 years in advance in 5-year chunks. EV charging infrastructure deliberately specified during terminal construction.	High
Is the fire truck battery compromised for short-distance use?	No. Battery capacity is sized for sustained firefighting operations, not constrained by the short drive time to any point on the airport.	High
Can international waste be sorted before sterilisation?	In principle, yes. Airside is a controlled biosecurity zone under MPI jurisdiction. Sorting quarantine waste by material type before steam sterilisation at Wigram could enable post-sterilisation material recovery, but would require a biosecurity-compliant sorting protocol and MPI approval. Not yet implemented.	Medium
What is the airport's emission scope boundary?	Scope 1 and 2 only for the "absolute zero by 2035" target. Scope 3 (airline flights, ~750,000 tCO_2e) is measured and publicly reported but separate.	High

WHAT WE ASKED	WHAT WE FOUND	CONFIDENCE
What happens to baggage tags at end of journey?	Baggage tags represent significant single-use plastic and adhesive waste across millions of passenger journeys. Alternatives exist, including digital tags and paper-only formats, but no airport-wide standard has emerged. This is a question for airlines as much as airports.	Low-Medium

Open Questions and Next Steps

These are questions that arose during the tour and could not be fully resolved through post-tour research. They represent opportunities for CIAL and the wider group to lead on.

Cleaning contract scope: What would need to change in CIAL's facilities management arrangements to support a centralised reusable serviceware programme? This is a question for Shannon's team to scope internally before it becomes a public recommendation.

Confiscated item reuse: Can functional items confiscated at security (power banks, electronics) be legally redistributed or resold, or does aviation security law prevent this?

Ground power health data: Has anyone measured the health benefit to ground crew from reduced exposure to jet fuel fumes (PAH) since the switch to electric ground power? This data would strengthen the business case for other airports.

Quarantine waste sorting: What would an airside biosecurity sorting station look like, and what would MPI require to approve it?

Air bridge wheel recycling: Can Treadlite and Tyrewise confirm they will accept solid bonded industrial wheels with steel centres?

Participant Perspectives

This section draws from anonymised pre- and post-tour survey responses. No individual participants are identified.

What shifted

Before the tour, participants were broadly interested in understanding how a large organisation integrates sustainability into business-as-usual, how to build internal buy-in, and how to tell sustainability stories credibly. Most expected the energy transition to be the hard part and waste management to be more straightforward.

After the tour, this expectation was inverted. Several participants noted that providing clean electricity to parked planes was "easy", the infrastructure just needed to be built, while waste and circularity turned out to be the genuinely difficult challenge, entangled with human behaviour, tenant relationships, and commercial incentives.

The local reception environment was the single biggest surprise. Multiple participants expressed shock at the gap between CIAL's global standing and the hostility of local responses. As one participant wrote:

> *"I expected in-terminal passenger behaviour to be the biggest challenge, but community engagement in the journey was a bigger struggle."*

Honesty resonated

The strongest recurring theme was appreciation for the tour's honesty. Participants valued hearing about failures (the autonomous shuttle with no use case, the "climate positive" story, the Avenex grass reassessment) as much as successes. One participant wrote:

> *"It takes more courage to step into that light knowing you're doing the right thing for people and planet, than to withdraw and be small."*

A shared challenge

Several participants reported experiencing similar communications backlash in their own organisations, including Commerce Commission complaints about sustainability claims, social media hostility, and the political weaponisation of "virtue signalling." The tour created a sense of solidarity:

> *"We are all dealing with similar challenges regardless of organisation size. Supporting and advocating for each other is so important."*

Business case recalibration

An interesting pattern emerged in the survey data. Several participants who assessed themselves as having a "strong business case" or "some wins" before the tour adjusted their self-assessment downward afterward, moving to "know it matters but have not cracked the commercial argument yet." This is a healthy recalibration. The tour revealed how complex sustainability-business integration really is at operational scale. Seeing the depth of CIAL's work, and the challenges that persist even with board support, long-term investment, and deep technical capability, raised the bar for what a genuine sustainability business case looks like. This is not discouragement; it is a more accurate understanding of the journey.

One underreported dimension is how CIAL's sustainability investments have quietly reduced the airport's operational risk profile. The electric fire truck became a resilience asset during New Zealand's diesel supply constraints, a benefit not anticipated at purchase and not currently being systematically measured. The same pattern likely applies across other electrification decisions. This is a story about risk management as much as sustainability, and it is not yet being told that way.

Recommendations

Quick wins (weeks to months)

1. Develop a tenant adoption strategy for reusable serviceware, anchored in an accusation audit and resistance map. Before approaching SSP and other food court operators, identify every likely objection and prepare responses using the 5 Spheres of Empathy prompt chain (free resource), available at https://5spheresofempathy.com/resources, to map resistance across direct and indirect stakeholders. Then initiate conversations with willing tenants first and build a phased rollout from demonstrated success. This effort takes approximately 2 hours in full with more time invested in the empathy map to gain stronger outputs for the strategy.

2. Test transparent bins in one food court zone. Low cost, low risk. Run for 4 weeks alongside existing Method bins. Measure contamination rate before and after using the existing BI tablet system.

3. Test push-lid bins in one food court zone. Same approach: parallel trial, measured through existing data infrastructure. Mitchell Graham has offered to share insights from his own trials.

4. Contact Trelleborg about air bridge wheel take-back. A single email or phone call to the OEM supplier could resolve the wheel recycling question. Simultaneously contact Treadlite (Cambridge, NZ) and Tyrewise to confirm scope.

5 When the Flybrary is ready to launch publicly, it is a story worth supporting. It is exactly the kind of honest, human-scale initiative that resonates with the public, a first-in-New Zealand airport library stocked entirely with books salvaged from the waste stream. One consideration: CIAL's relationship with Relay, their bookshop tenant, should be factored into the Flybrary rollout plan. The library should complement rather than compete with an existing commercial partner.

Medium-term (months to a year)

6 Pilot a PDX-style Green Plate program with one food court tenant. Start with the SSP outlet most aligned with dine-in service. Airport owns the plates. Cleaning contractor washes. Default is reusable. Takeaway packaging on request only. Measure waste reduction against baseline. Reusable serviceware systems of this kind have operated sustainably at food courts globally, from Singapore to Poland, for decades. This is not an experimental model. It is an established one that airports have been slow to adopt.

7 Scope the facilities management arrangements needed to support centralised dishwashing. CIAL has identified a space adjacent to the food court that could serve this function. Shannon's team is the natural internal owner of this scoping work. The question is organisational: what fit-out would the space require, and how would a centralised washing function integrate with the current facilities management setup? A secondary question worth asking in that scoping process: what would this framework need to look like for other venue and facilities operators to adopt similar arrangements into their future contracts? CIAL has the scale and credibility to set a replicable model, not just solve its own problem.

8 Explore airside biosecurity sorting station. The "sort before sterilisation" idea has genuine potential. Begin with a desktop scoping study: what does MPI require, what waste streams would be separated, what are the cost savings from reduced landfill volume? Note that biosecurity waste is among the most expensive waste streams at any airport. It is processed through an autoclave before specialist burial at landfill. Any diversion upstream of that process carries significant cost saving potential, making this worth prioritising.

9 Develop a one-page GPU business case for peer airports. CIAL's ground power data (cost, energy savings, emissions reduction) could accelerate adoption elsewhere. This is low effort, high impact knowledge sharing.

Longer-term (year-plus)

10 Airside heat loss through vehicle openings.Commission a thermal audit of airside vehicle access points to quantify infiltration losses and evaluate intervention options including air curtains, high-speed doors, and destratification systems. Peer-reviewed research across airport terminals has documented infiltration accounting for up to 71% of total heat loss, with proven reductions of over 80% achievable through envelope improvements

11 Confirm REVYRE capability for solid rubber recycling. REVYRE appears to already have a presence in North Canterbury. Confirming their current capability and capacity for solid rubber recycling should be treated as a near-term action rather than a longer-term watch item.

12 Build a Christchurch sustainability communications peer support network. The tour revealed that multiple organisations are independently experiencing the same local backlash. A small, informal group that shares communications strategies and provides mutual support would cost nothing and could be meaningfully protective. A first step would be identifying three or four organisations from this tour group willing to anchor the network and agree on a format. Several participants from today's group would be natural candidates.

This report was produced by Sustainability Tours from tour observations, participant survey responses, Mural board contributions, and post-tour research. It reflects the collective experience of the group, not the views of any single participant or organisation.

Sustainability Tours | mr.g@theideaconsultants.com

Photography by Kai J. Lee — creative storyteller and sustainability documentarian. Kai joined this tour as a participant and captured it with a generosity that made this report possible. See more of his work at subtledream.com